Andrea Capone

PhD Researcher in AI | Software Developer

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Profile

PhD Researcher in Artificial Intelligence with a strong background in software engineering. I work at the intersection of **Machine Learning**, **Explainable AI** and **formal methods**, applying neurosymbolic approaches to environmental risk assessment. Author of several publications and an active contributor to research software, with hands-on experience building full-stack and data-intensive applications.

Technical Skills

Languages: Python, Java, C

AI & ML: Machine Learning, Explainable AI, Neurosymbolic AI, Multi-Agent Systems, Formal Methods & Model Checking

Data & Databases: Neo4J, Firebase, Time-Series Analysis (GNSS), Data Analysis

Other: Full-Stack Development, Cloud Computing

Publications

- *Reasoning About Intuitionistic Alternating-Time Temporal Logic*. In Proceedings of the **8th International Workshop on Artificial Intelligence and fOrmal VERification, Logic, Automata, and sYnthesis (OVERLAY)**, 2026.
- *GameL: a Neuro-Symbolic Agentic Architecture Toward Autonomous and Resilient Energy Communities*. In Proceedings of the **41st Italian Conference on Computational Logic (CILC)**, 2026.
- *A Neurosymbolic Approach for Landslide Risk Assessment*. In Proceedings of the **6th National Conference on Artificial Intelligence (Ital-IA)**, 2026.
- *Gaussian Process and Kalman Baselines for Bradyseismic Signal Extraction from GNSS Time Series at Campi Flegrei*. In Proceedings of the **XXXIV Italian Workshop on Neural Networks (WIRN)**, 2026.
- *Machine Learning and Explainable Deep Learning for Forecasting Landslide Deformation using EGMS Time Series*. Preprint, **Applied Soft Computing**, 2026.
- *An Intuitionistic Version of Computation Tree Logic*. In Proceedings of the **22nd European Conference on Multi-Agent Systems (EUMAS)**, 2025.
- *An Intuitionistic Version of Alternating Time-Temporal Logic*. In Proceedings of the **22nd International Conference on Principles of Knowledge Representation and Reasoning (KR)**, 2025.

Projects

PhD Project: Neurosymbolic AI for Bradyseism Risk Assessment — Nov 2024 – Present
National PhD in AI for Agrifood and Environment — *Python, Machine Learning, Formal Methods*

- Extracting the bradyseismic ground-deformation signal from **GNSS time series** at **Campi Flegrei**, comparing **FastICA**, **Gaussian Process**, and **Kalman-filter** baselines, with FastICA providing the most physically meaningful source separation
- Building robust processing pipelines centered on **outlier** treatment and recovery of the **deterministic, nonlinear deformation trend**, yielding clean and interpretable signals
- Investigating **neurosymbolic** approaches that ground **Explainable AI** outputs and neural predictions into symbolic representations, enabling formal reasoning to validate model outcomes and filter unreliable predictions
- Collaborating with **INGV** to turn these analyses into decision-support tools for risk mitigation and resilience strategies

APLAND — 2025 – 2026
AI for Landslide Risk Assessment — *Python, FastAPI, Streamlit, Docker*

- Developed a **Frequency Ratio** algorithm for landslide susceptibility, served through a **FastAPI** backend with a **Streamlit** interface and a **Docker** setup simulating a **GeoServer** environment
- Supervised the development of a temporal neural pipeline for **explainable forecasting** of **EGMS** deformation time series, combined with **neural spatial susceptibility mapping**
- Developed a **decoupled neurosymbolic framework** for landslide risk assessment, published at **Ital-IA 2026**

Medical Survey System — Feb 2025 – Present
Interactive Surveys for Medical Conferences — *JavaScript, Firebase*

- Built an **interactive survey platform** for medical conferences, enabling real-time data collection and live visualization
- Integrated **Firebase** for secure authentication, data storage, and real-time synchronization
- Developed responsive **JavaScript** front-end components ensuring smooth usability across devices

VITAMIN Model Checker — Feb 2024 – Present
Logic for Computer Science, Multi-Agent Systems — *Python*

- Active contributor to **VITAMIN**, a state-of-the-art model checker for multi-agent systems
- Designed and implemented model-checking procedures for **Intuitionistic Computation Tree Logic (ICTL)** and **Intuitionistic Alternating-time Temporal Logic (IATL)**
- Optimized core routines and data-structure management, significantly improving scalability and performance

Food Talk — Jun 2023 – Present
Sustainability, Software Development — *Python, Neo4J*

- Designed and developed a **Neo4J**-based system for end-to-end traceability of fruit and vegetable products
- Modeled highly interconnected supply-chain relationships as a graph for efficient querying and analysis
- Exploring partnerships with industry and the University of Naples to scale the platform and advance research

Theater Section — Jun 2019 – Feb 2020
Human-Computer Interaction, Software Development — *Java, Android, Neo4J*

- Developed a system addressing the exhibition and conservation needs of the **“Fondo Niccolini”** collection
- Used **Neo4J** to model the highly interconnected and complex relationships within the archival data
- Deployed and currently in operation at the **“Certosa di San Martino”**, Naples

Experience

Research and Development Internship — Feb 2023 – Nov 2023
Maipek s.r.l. | University Federico II, Naples, Italy

- Conducted R&D on an **intraoral scanner**, focusing on 3D reconstruction from optical scan data
- Identified and benchmarked image-processing techniques to build accurate and effective **point clouds**

Education

Ph.D. in Artificial Intelligence for Agrifood and Environment — **Status:** Ongoing
National PhD Program in AI | University Federico II, Naples, Italy — *Nov 2024 – Present*

M.Sc. in Computer Science, curriculum in Reliable Software Systems — **Grade:** 110/110 Cum Laude
University Federico II, Naples, Italy — *Sep 2020 – Jul 2024*

B.Sc. in Computer Science — **Grade:** 94/110
University Federico II, Naples, Italy — *Sep 2014 – Mar 2020*

Certifications

- Apple Developer Academy
- Agritech Student Academy

Languages

- Italian – Native Speaker
- English – Professional Working Proficiency

Agreements

I authorize the processing of the personal data contained in my CV pursuant to Article 13 of Legislative Decree 196/2003 and Article 13 of the GDPR (EU Regulation 2016/679).